Gap-Filling of Highly Incomplete Daily Chlorophyll-a Remote Sensing Time-Series Data Over the Eastern China Seas via Spatiotemporal-Periodicity Aware Tensor Completion

Gaoxiang Zhou, Chao Ding, Zeifei Ming, Ming Xu, and Ming Liu

***Abstract*—The application of daily Chlorophyll-a (Chla) concentration remote sensing data is inevitably affected by large gaps caused by clouds, sunlight, and other undesirable conditions. Existing methods face challenges in addressing large data gaps due to their limited ability to leverage spatiotemporal correlation and periodic features. This study presents a spatiotemporal–periodicity aware tensor completion (STPTC) method for gap-filling highly incomplete daily Chla datasets over the eastern China seas. The method first performs periodic-aware tensor restructuring to align interannual seasonal patterns, then embeds spatial and temporal features through graph Laplacian, fractional temporal differencing, and periodic circulant transforms, followed by adaptive low-rank tensor completion with mode-wise weight optimization. Experiments using ten years of Moderate Resolution Imaging Spectroradiometer (MODIS)-Aqua daily Chla data over the eastern China seas, with an average missing rate of 87%, demonstrate that the proposed method consistently achieves high reconstruction accuracy across multiple years, with correlation coefficients (CCs) above 0.97, root mean square differences (RMSDs) below 0.43 mg/m^3, and mean absolute percentage differences (MAPDs) under 15.2%. The method remains robust even under missing rates up to 95%. Compared with linear interpolation, Data Interpolating Empirical Orthogonal Functions (DINEOF), DINCAE, CP, and Tucker, the proposed method achieves the highest overall accuracy and superior spatial–temporal consistency, particularly under complex missing patterns. Additionally, ablation results confirm the complementary contributions of the proposed spatial, temporal, and periodic transform operations. The findings indicate that the proposed method provides a promising approach for filling large data gaps in the daily Chla remote sensing dataset, which could enhance the application of Chla data in marine ecosystem-related studies.**

***Index Terms*—Daily chlorophyll-a (Chla) remote sensing data, gap-filling, highly incomplete, spatiotemporal-periodicity, tensor completion.**

I. Introduction

CHLOROPHYLL-a (Chla) concentration serves as a crucial measure of phytoplankton status and biomass, commonly used in marine ecosystem productivity assessments, phytoplankton phenology detection, and algae bloom monitoring [1], [2], [3], [4]. Satellite remote sensing has become a dominant approach for tracking Chla changes worldwide, offering extensive coverage of large areas with temporal resolutions ranging from daily to monthly [5]. However, unfavorable conditions such as cloud cover and sunlight interference result in inevitable gaps in the Chla data products obtained from satellites, hindering their application in phytoplankton-related studies [6], [7]. For example, over 70% of global daily ocean color data from Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS) is missing [8]. Therefore, filling the missing data gap in the Chla time series is crucial for accurately capturing changes in marine phytoplankton across extensive spatial and temporal scales.

Many gap-filling methods have been proposed to reconstruct missing information in Chla time-series products. These can be broadly categorized into traditional interpolation methods and learning-based methods. Traditional interpolation methods exploit the inherent spatial and temporal dependencies of Chla itself without external data [9]. Early methods such as linear and optimal interpolation are straightforward and have strong generalization capabilities, making them easy to use for marine remote sensing data recovery [10], [11]. However, their reliance on the temporal information of single pixels or the spatial information of images limits their effectiveness,

Received 2 September 2025; revised 26 October 2025, 29 December 2025, and 22 February 2026; accepted 23 February 2026. Date of publication 26 February 2026; date of current version 9 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42206175; and in part by the Fundamental Research Funds for the Central Universities under Grant 300102354201 and Grant 2452025020. *(Corresponding authors: Gaoxiang Zhou; Ming Liu.)*

Gaoxiang Zhou is with the State Key Laboratory of Soil and Water Conservation and Desertification Control and the College of Soil and Water Conservation Science and Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China (e-mail: zgx403318@gmail.com).

Chao Ding is with the Department of Geographic Science, Faculty of Arts and Sciences, Beijing Normal University, Zhuhai 519087, China.

Zeifei Ming and Ming Xu are with the Faculty of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China.

Ming Liu is with the School of Land Engineering, Chang'an University, Xi'an 710064, China (e-mail: mingliu@chd.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3668244, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3668244

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

especially when large gaps in time or space are present [12]. To address this issue, spatiotemporal interpolation methods have been proposed. Among these, the Data Interpolating Empirical Orthogonal Functions (DINEOF) method is the most representative, which reconstructs missing data based on empirical orthogonal functions and has become the most widely used apprSta-gan: A spatiotemporal1007 attention generative adversarial network for missing value imputation in1008 satellite data,oach for filling missing data in ocean remote sensing data [13], [14]. Many studies have confirmed its effectiveness in reconstructing ocean color or other ocean environmental variables [8], [15], [16]. By extracting leading modes, DINEOF captures large-scale spatiotemporal variability robustly. However, because it depends mainly on these dominant patterns, its performance may decline in highly dynamic ocean areas or under extensive gaps, where small-scale and short-term variations are less well represented [17]. In addition, some researchers proposed a spatiotemporal interpolation method to fill gaps in 8-day composite Chla products using spatially adjacent pixels with similar temporal variations, but this method cannot be applied to datasets with high missing rates [18].

Learning-based methods aim to establish empirical relationships from data, either using Chla time series alone or integrating auxiliary environmental drivers. Early studies employed machine learning, such as the self-organizing maps model [19] and random forest algorithms [20], for data gap filling. More recently, deep learning (DL) methods have been developed, including the Data Interpolating Convolutional Autoencoder (DINCAE) networks [21], [22], *Convolutional Neural Networks (CNNs)* [23], and Attention Network [24]. Newer models extend these ideas to global scales and near-real-time applications [25], [26]. Although these methods have achieved promising accuracy, task-specific DL models are still limited by the size of the training dataset, spatiotemporal diversity, and fixed model parameters [9]. For instance, the reliance on large and representative training datasets will hinder performance in cases where such data is scarce or costly to obtain. Fixed model parameters may not adapt well to evolving spatiotemporal dynamics, potentially leading to suboptimal performance when the underlying conditions deviate from the training environment. For global-scale applications, learning-based gap-filling methods often operate at low spatial resolution to simplify model construction and reduce running costs, which limits their ability to characterize detailed variation [23]. Recently, advanced learning-based frameworks have been introduced to further enhance ocean color data reconstruction. For example, an AI-powered Spatiotemporal Imputation and Prediction Model (STIMP) that combines Transformer and diffusion networks with a spatial graph module to reconstruct missing data and predict coastal short-term Chla variations was proposed in [27]. The model effectively captures temporal variability and spatial differences in complex coastal environments. In addition, a foundation model for ocean color remote sensing, referred to as the Ocean Color Foundation Model (OCFM), was presented in [28]. It integrates satellite observations, derived products, and in situ datasets through large-scale pretraining and task-specific fine-tuning to enable unified inversion of multiple ocean optical parameters and cross-region generalization. These studies represent the evolution from task-specific deep networks toward generative and foundation-level models that emphasize transferability, scalability, and multitask learning. Despite these advances, such data-driven advanced models still require large amounts of training data, strong computing resources, and relatively complete time series, which are often difficult for daily Chla products with very high missing rates.

In this context, traditional interpolation methods remain an efficient and complementary approach that directly exploits the spatiotemporal information from the data itself to fill gaps in the Chla time series, which makes them the most widely used methods [16], [29], [30], [31], [32]. However, these methods still do not fully utilize temporal and spatial correlation information. Temporally, existing interpolation methods rely solely on temporally adjacent observations and ignore the periodic pattern of phytoplankton growth [33]. Similarly, current spatiotemporal methods are insufficiently used for this periodicity feature and do not take into account the differences in the contribution of different spatiotemporal dimensions to the recovery of missing data [16], [34]. Indeed, long-term Chla time series accumulated over decades can be mined for information in different dimensions, such as the spatial domain, temporal domain, and annual periodic patterns, to fill in missing data. However, current methods lack an effective strategy for simultaneously integrating multidimensional information to recover missing data in Chla time-series products. These limitations constrain the applicability of existing methods, particularly in regions with high rates of missing data [9], [18].

As a typical form of spatiotemporal data, remote sensing time series can be naturally represented as a 3-D tensor, with rows, columns, and time as the respective spatial and temporal dimensions. Furthermore, due to the spatial and temporal similarities within the data, remote sensing time series exhibit low-rank tensor characteristics [35]. The stronger this low-rank property, the greater the spatiotemporal correlation within the data. Consequently, low-rank tensor completion methods can be applied to recover missing data in such low-rank tensors. This approach has already been explored with promising results in the reconstruction of missing data in multispectral images and terrestrial remote sensing vegetation indices [36], [37]. Chla time-series data, influenced by the growth patterns of phytoplankton, also exhibit strong spatiotemporal similarities. Consequently, their time-series tensors are well-suited for low-rank tensor completion methods to recover missing data. Nevertheless, compared with terrestrial remote sensing applications, the missing rate in daily Chla data is substantially higher (often exceeding 80%), posing a much greater challenge for accurate reconstruction. This difficulty arises not only from the severe data missing but also from the fact that most existing tensor completion approaches primarily focus on low-rank structure recovery without explicitly modeling the ecological priors underlying marine Chla dynamics. In particular, they rarely account for spatial correlation, temporal continuity, and long-term periodicity simultaneously, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

are key to accurately reconstructing daily Chla time series under severe missingness.

Broadly, two main strategies have been developed for low-rank tensor completion, including factorization-based approaches and nuclear norm minimization [38]. Factorization-based approaches, such as CANDECOMP/PARAFAC (CP) and Tucker decomposition [39], [40], directly decompose the tensor into low-rank factors. These methods are computationally efficient but nonconvex, and their accuracy often degrades under irregular missing patterns or spatial heterogeneity, both of which are common in Chla remote sensing data. Moreover, they do not explicitly incorporate domain-specific spatiotemporal priors, such as spatial correlation, temporal continuity, or seasonal periodicity, which are essential for improving tensor completion performance under scenarios with extensive missing data [41]. In contrast, nuclear norm minimization provides a convex relaxation of the rank function, offering theoretical guarantees of global optimality. Although more computationally demanding, it is generally more robust under severe missingness and can be flexibly extended to incorporate additional prior constraints. Therefore, this study builds upon the nuclear norm minimization framework and enhances it by embedding spatiotemporal and periodic priors, forming a unified tensor completion model tailored to the characteristics of daily Chla time series.

Building on this foundation, we develop a Spatiotemporal–Periodicity Aware Tensor Completion (STPTC) model that jointly leverages spatial, temporal, and seasonal priors within an integrated low-rank optimization framework. Unlike conventional tensor completion approaches that rely solely on low-rank structure, the proposed STPTC model explicitly integrates domain-relevant priors, including spatial correlation (via Graph Laplacian), temporal consistency (via fractional differencing), and annual periodic recurrence (via periodic circulant transform), into an adaptive nuclear norm minimization framework. An adaptive weighting strategy is further embedded in the Alternating Direction Method of Multipliers (ADMM) optimization, allowing the model to dynamically balance the contributions of each spatiotemporal prior based on their unfolding singular value structures. This integration not only ensures robust reconstruction of highly incomplete daily Chla datasets but also maintains transferability to other marine variables such as sea surface temperature and particulate organic carbon. The resulting gap-filled Chla could provide a valuable data foundation for subsequent applications, including phytoplankton bloom monitoring, ecosystem productivity analysis, and climate-driven variability assessment.

II. Methodology

To address the challenge of reconstructing severely missing remote sensing Chla time series, we propose a three-stage tensor completion method that integrates prior knowledge from spatial, temporal, and periodic perspectives. As illustrated in Fig. 1, the method consists of three key components: 1) *periodic-aware tensor restructuring*, which reorganizes the original time series into a seasonally aligned format to better capture inter-annual periodicity and low-rank structures; 2) *spatiotemporal feature embedding via transform operations*, where graph-based spatial smoothing, fractional temporal differencing, and interannual circulant constraints are applied to enhance structural consistency across dimensions; and 3) *adaptive low-rank tensor completion*, which exploits the enhanced multimode correlations with learnable mode-wise weights to recover missing values under severe data sparsity.

A. Periodic-Aware Tensor Restructuring

Traditionally, remote sensing time-series data can be represented as a 3-D tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times t}$, where m and n represent the number of rows and columns in an image, and t represents the number of temporal observations. While this tensor structure captures the spatiotemporal relationships between images, it does not effectively reflect the periodic variations inherent in time-series data, such as Chla. To better capture the seasonal periodicity of Chla time series and reveal structured low-rank patterns, we transform the original Chla tensor into a periodically aligned format [36].

The original Chla tensor was first unfolded along the temporal dimension to form a 2-D tensor $\mathcal{X}' \in \mathbb{R}^{(m\times n)\times t}$. Next, the temporal dimension was reorganized based on the number of years and the days within each year, yielding a 3-D tensor $\mathcal{X}'' \in \mathbb{R}^{(m\times n)\times n_y\times n_d}$, where n_y represents the number of years, and n_d represents the number of days within each year (e.g., 365 for MODIS daily composites). This rearrangement enables the model to align interannual seasonal cycles along the fourth dimension and leverage low-rank structures across space, intraannual time, and years.

B. Spatiotemporal Feature Embedding via Transform Operations

To further enhance low-rank structures and explicitly incorporate prior knowledge, we apply three domain-specific transforms to encode spatial, temporal, and periodic relationships.

1) Graph Laplacian Transform Matrix: The Graph Laplacian matrix $\mathbf{T}_1 \in \mathbb{R}^{n_1\times n_1}$ encodes spatial local consistency by constructing an adjacency-based representation of pixel relationships. It is defined as

$$\mathbf{T}_1 \triangleq \mathbf{T}_W - \mathbf{W} \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{n_1\times n_1}$ represents a weighted adjacency matrix, and $\mathbf{T}_W \in \mathbb{R}^{n_1\times n_1}$ is a diagonal degree matrix with diagonal elements

$$(\mathbf{T}_W)_{ii} = \sum_{j=1}^{n_1} W_{ij}, \quad i = 1, \ldots, n_1. \tag{2}$$

In the context of the Chla tensor, the weighted adjacency matrix for the pixels is derived from the distances between pixels

$$W_{ij} = \begin{cases} \exp\left(-\dfrac{d_{ij}^2}{\sigma^2}\right), & i \neq j \text{ and } \exp\left(-\dfrac{d_{ij}^2}{\sigma^2}\right) \geq \epsilon \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where d_{ij} denotes the distance between pixels, σ^2 regulates the distribution of the adjacency matrix, and $\epsilon > 0$ serves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

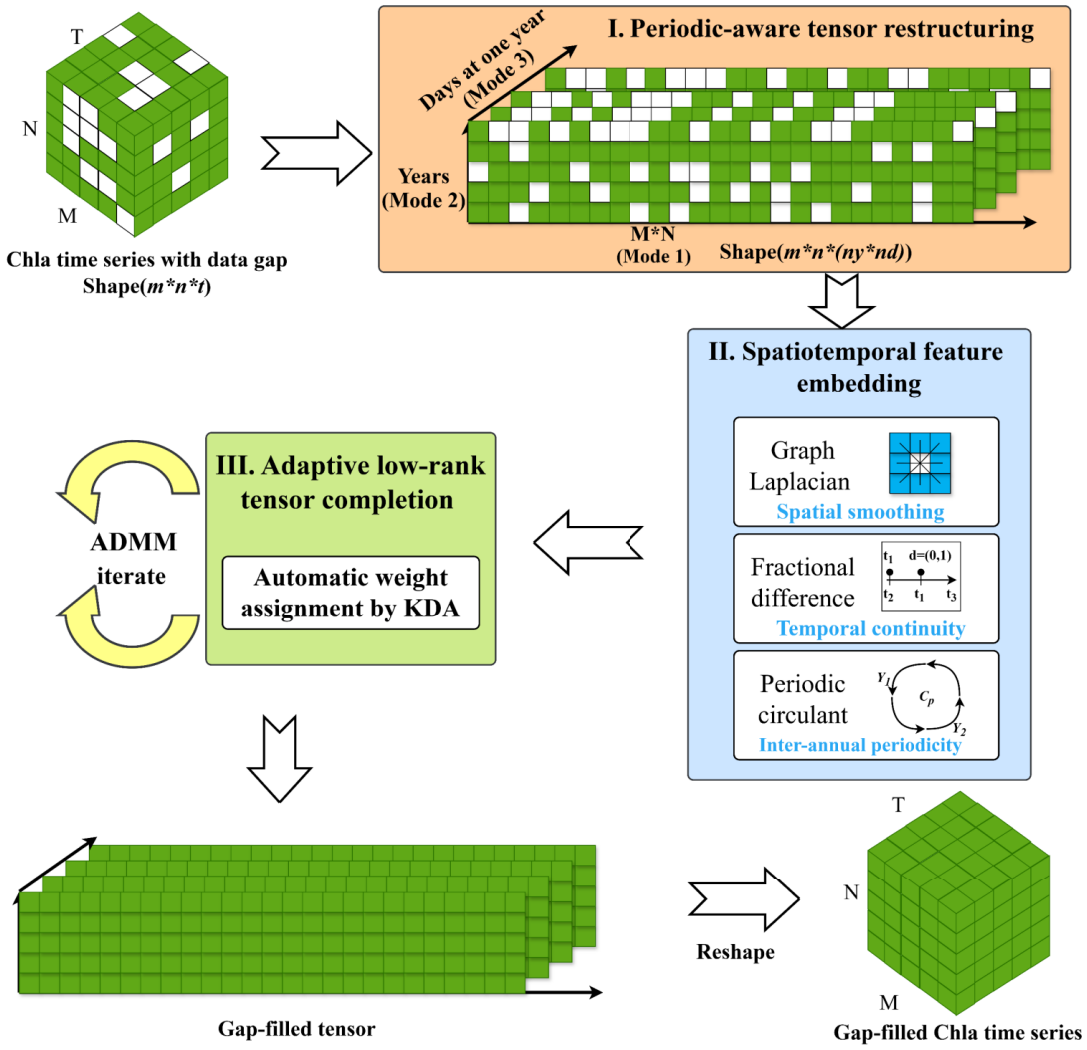


Fig. 1. Flowchart of the proposed method.

to truncate smaller values. The application of the Graph Laplacian transformation to the feature tensor implies that Chla data from pixels situated close to each other at the same year on the same day will yield similar values.

2) Fractional Difference Transform Matrix: The fractional difference matrix $\mathbf{T}_2 \in \mathbb{R}^{n_2 \times n_2}$ is designed to capture temporal consistency by modeling smooth transitions over time while retaining long-term memory. It is constructed based on the discrete Grünwald–Letnikov (GL) fractional derivative [37]. For any temporal vector $\mathbf{x} = [x_1, \ldots, x_{n_2}]^{\mathrm{T}}$, the discrete fractional-order difference is defined as

$$(\nabla^{\alpha} x)_i \triangleq \sum_{k=0}^{K-1} g_k \, x_{i-k}, \quad i = 1, \ldots, n_2 \tag{4}$$

where $x_i = 0$ for $i < 1$ or $i > n_2$, and the coefficients are given by

$$g_k = (-1)^k \binom{\alpha}{k}, \qquad \binom{\alpha}{k} = \frac{\Gamma(\alpha+1)}{\Gamma(k+1)\Gamma(\alpha-k+1)} \tag{5}$$

with $\Gamma(\cdot)$ being the Gamma function.

Accordingly, $\mathbf{T}_2$ can be explicitly written as a lower-triangular Toeplitz matrix

$$\mathbf{T}_2 \triangleq \begin{bmatrix} g_0 & 0 & 0 & \cdots & 0 \\ g_1 & g_0 & 0 & \cdots & 0 \\ g_2 & g_1 & g_0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{n_2-1} & g_{n_2-2} & g_{n_2-3} & \cdots & g_0 \end{bmatrix}_{n_2 \times n_2}. \tag{6}$$

Thus, applying $\mathbf{T}_2$ to $\mathbf{x}$ is equivalent to computing the fractional difference: $\mathbf{T}_2\mathbf{x} = \nabla^{\alpha}\mathbf{x}$.

Special cases include: 1) when $\alpha = 1$, Eq. (4) reduces to the standard first-order difference and 2) when $0 < \alpha < 1$, the operator represents a fractional difference that incorporates contributions from multiple past observations with geometrically decaying weights. For example, with $\alpha = 0.5$, the first few coefficients are $[1, -0.5, -0.125, \ldots]$, meaning current values are influenced not only by the immediate past but also by longer historical terms with gradually decreasing influence.

This property allows fractional differencing to simultaneously smooth short-term variations and preserve long-term memory in the Chla time series, making $\mathbf{T}_2$ well-suited for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

modeling oceanographic processes that exhibit both seasonal cycles and interannual persistence.

3) Periodic Circulant Transform Matrix: The periodic circulant matrix $\mathbf{T}_3 \in \mathbb{R}^{n_3 \times n_3}$ is introduced to explicitly encode the annual periodicity of phytoplankton growth. By construction, it enforces similarity between Chla values separated by one seasonal cycle, thereby reinforcing interannual consistency in the reconstructed series [38].

Formally, when the Chla time series follows a periodic pattern with cycle length t (e.g., 365 days for daily MODIS composites), $\mathbf{T}_3$ is defined as a circulant matrix generated by the vector

$$\mathbf{g} = [-1, 0, \ldots, 0, 1, 0, \ldots, 0] \tag{7}$$

where the first entry is $g_1 = -1$, the $(t+1)$th entry is $g_{t+1} = 1$, and all other entries are zero. This structure enforces that values separated by exactly one period contribute with opposite signs, thereby regulating periodic similarity across the time axis. For example, when $t = 2$ and $n_3 = 5$, the periodic circulant matrix takes the form

$$\mathbf{T}_3 = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 \end{bmatrix}. \tag{8}$$

It is worth noting that the annual periodicity considered in this study corresponds to a 365-day climatological cycle. Therefore, a 366-day periodicity is not assumed for leap years. During the construction of the periodically restructured tensor, the leap-year daily series is mapped onto a unified 365-day DOY grid through a calendar-aware, local interpolation-based treatment around February 29. In this way, the subsequent periodic operator remains consistently defined and applied on DOY $= 1, \ldots, 365$. The detailed implementation, including the explicit redistribution and recovery expressions, is provided in the Supplementary Material.

In practice, applying $\mathbf{T}_3$ to the Chla tensor ensures that observations from the same day in different years are strongly coupled, which is essential for recovering missing data under high sparsity. This periodic constraint is particularly valuable because phytoplankton dynamics are dominated by seasonal bloom cycles. As a result, $\mathbf{T}_3$ complements the spatial and temporal priors by embedding annual periodicity directly into the tensor completion process.

The resulting spatiotemporal feature-enhanced tensor $\mathcal{H}_k$ is then obtained via the mode-k product

$$\mathcal{H}_k \triangleq \mathcal{X}'' \times_k \mathbf{T}_k \tag{9}$$

where $\mathcal{X}''$ is the periodically restructured tensor and $\mathbf{T}_k$ denotes the kth transform matrix ($k = 1, 2, 3$). In this way, spatial adjacency, temporal smoothness, and annual periodicity are explicitly embedded into the tensor representation. The feature-enhanced tensor thus incorporates complementary priors across multiple domains, providing a more structured and correlated representation of the Chla series.

C. Adaptive Low-Rank Tensor Completion

When the tensor has a relatively low rank, the recovery of missing data can be formulated as a low-rank tensor completion task using a partially observed data tensor [35]. Consequently, the missing data in the natural low-rank Chla time-series tensor can be filled by low-rank tensor completion methods. In this study, we formulate the problem of recovering missing data in the Chla tensor after spatiotemporal feature enhancement $\mathcal{H}_0$ as

$$\min_{\mathcal{H}} \ \operatorname{rank}(\mathcal{H}), \quad \text{s.t.} \quad P_\Omega(\mathcal{H}) = P_\Omega(\mathcal{H}_0) \tag{10}$$

where rank$(\cdot)$ represents the tensor algebraic rank of the Chla time series. Ω denotes the index set of the valid data in the tensor, and P_Ω is the projection operator that keeps entries in Ω and sets the others to zero

$$[P_\Omega(\mathcal{H})]_{\text{index}} = \begin{cases} H_{\text{index}}, & \text{if index} \in \Omega \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Equation (8) represents a typical rank minimization problem, where the primary challenge lies in approximating the rank of the tensor [42]. The core idea behind low-rank completion is to recover missing values by imposing a low-rank constraint, which encourages the tensor to capture underlying patterns and relationships among the data. However, defining and calculating the rank of a tensor is complex, as it often depends on correlations across multiple dimensions [43]. Directly minimizing the rank is generally NP-hard and computationally impractical. As a result, alternative approaches have been proposed. One of the most popular involves approximating the rank of the tensor using the nuclear norm, which is the sum of its singular values and serves as a convex relaxation of the rank function [44]. This technique effectively promotes low-rank structure while being computationally feasible.

To make the optimization problem tractable, we adopt a nuclear norm relaxation, replacing rank minimization with a convex surrogate

$$\min_{\mathcal{H}} \ \sum_{k=1}^{3} \alpha_k \left\| \mathcal{H}_{(k)} \right\|_*, \quad \text{s.t.} \quad P_\Omega(\mathcal{H}) = P_\Omega(\mathcal{H}_0) \tag{12}$$

where $\mathcal{H}_{(k)}$ denotes the unfolding of $\mathcal{H}$ along the kth mode, and $\|\cdot\|_*$ indicates the nuclear norm. The weights α_k are nonnegative and satisfy $\sum_{k=1}^{n} \alpha_k = 1$.

The weights α_k determine the contribution of each unfolding matrix in (10). To adaptively assign these weights, we analyze the singular value distributions of each unfolding. Let

$$\hat{l}_k = \frac{i_k}{n_k} \tag{13}$$

where i_k is the number of dominant singular values identified by the knee detection algorithm (KDA) [45], and n_k is the total number of singular values in the kth unfolding. The metric $\hat{l}_k$ quantifies the low-rankness of the kth unfolding via its singular value decay: a smaller $\hat{l}_k$ indicates faster decay and stronger low-rank property, whereas a larger $\hat{l}_k$ implies weaker low-rankness (see Fig. S1). Accordingly, α_k should be smaller for dimensions with stronger low-rankness to avoid over-penalizing dominant structures, and larger for dimensions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

with weaker low-rankness to enforce stronger regularization. We therefore define

$$\alpha_k = \frac{1/\hat{l}_k}{\sum_{j=1}^{n} 1/\hat{l}_j} \tag{14}$$

ensuring that dimensions with stronger low-rank structures (smaller $\hat{l}_k$) are assigned smaller penalties, while dimensions with weaker low-rank structures receive stronger regularization. The choice $\alpha_k \propto 1/\hat{l}_k$ preserves dominant low-rank components while imposing stronger constraints on weaker ones, achieving a balance between structural preservation and effective recovery. In practice, during each iteration of the ADMM optimization, α_k is updated dynamically according to this criterion.

D. Optimization via ADMM Algorithm

The ADMM algorithm is a widely used optimization method for solving tensor completion problems [46], [47], [48]. Thus, we adopt this algorithm to optimize (10). To enforce consistency between observed and reconstructed entries, we introduce an auxiliary variable $\mathcal{K}$ for the missing entries, constrained within the complement of the observed index set $\Omega^{\perp}$. This is expressed via the indicator function

$$\delta_{\mathcal{K},\Omega} = \begin{cases} 0, & P_{\Omega}(\mathcal{K}) = \mathcal{O} \\ +\infty, & \text{otherwise} \end{cases} \tag{15}$$

where Ω denotes the index set of observed entries, $\Omega^{\perp}$ denotes its complement (unobserved entries), and $\mathcal{O}$ is the zero tensor.

Accordingly, (10) can be reformulated as

$$\begin{aligned} \min_{\mathcal{H},\{\mathcal{H}_{(k)}\},\mathcal{K}} \quad & \sum_{k=1}^{3} \alpha_k \left\| \mathcal{H}_{(k)} \right\|_* + \delta_{\mathcal{K},\Omega} \\ \text{s.t.} \quad & \mathcal{H}_{(k)} = \text{unfold}_k(\mathcal{H}), \quad \mathcal{H} + \mathcal{K} = P_{\Omega}(\mathcal{H}_0) \end{aligned} \tag{16}$$

where $\mathcal{H}$ is the recovered tensor, $\mathcal{H}_0$ is the observed incomplete tensor, and $\mathcal{H}_{(k)}$ denotes the unfolding of $\mathcal{H}$ along the kth mode.

The augmented Lagrangian function is then defined as

$$\begin{aligned} &\mathcal{L}\left(\mathcal{H}, \{\mathcal{H}_{(k)}\}, \mathcal{K}, \{\mathcal{M}_k\}, \mathcal{N}\right) \\ &= \sum_{k=1}^{3} \alpha_k \left\| \mathcal{H}_{(k)} \right\|_* + \delta_{\mathcal{K},\Omega} \\ &\quad + \frac{\mu_t}{2} \sum_{k=1}^{3} \left\| \text{unfold}_k(\mathcal{H}) - \mathcal{H}_{(k)} + \mathcal{M}_k/\mu_t \right\|_F^2 \\ &\quad + \frac{\mu_t}{2} \left\| P_{\Omega}(\mathcal{H}_0) - \mathcal{H} - \mathcal{K} + \mathcal{N}/\mu_t \right\|_F^2 + \mathcal{C} \end{aligned} \tag{17}$$

where $\mathcal{M}_k$ and $\mathcal{N}$ represent the Lagrange multipliers, $\mu_t > 0$ is a penalty parameter, and $\mathcal{C}$ denotes terms depending only on multipliers.

Based on (17), the ADMM framework solves the following subproblems iteratively

$$\mathcal{H}^{t+1} = \arg\min_{\mathcal{H}} L\left(\mathcal{H}, \mathcal{H}_{(k)}^{t}, \mathcal{K}^{t}, \mathcal{M}_k^{t}, \mathcal{N}^{t}\right) \tag{18}$$

$$\mathcal{H}_{(k)}^{t+1} = \arg\min_{\mathcal{H}_{(k)}} L\left(\mathcal{H}^{t+1}, \mathcal{H}_{(k)}, \mathcal{K}^{t}, \mathcal{M}_k^{t}, \mathcal{N}^{t}\right) \tag{19}$$

$$\mathcal{K}^{t+1} = \arg\min_{\mathcal{K}} L\left(\mathcal{H}^{t+1}, \mathcal{H}_{(k)}^{t+1}, \mathcal{K}, \mathcal{M}_k^{t}, \mathcal{N}^{t}\right). \tag{20}$$

The multipliers are updated as

$$\mathcal{M}_k^{t+1} = \mathcal{M}_k^{t} + \mu_t \left(\text{unfold}_k\left(\mathcal{H}^{t+1}\right) - \mathcal{H}_{(k)}^{t+1}\right) \tag{21}$$

$$\mathcal{N}^{t+1} = \mathcal{N}^{t} + \mu_t \left(P_{\Omega}(\mathcal{H}_0) - \mathcal{H}^{t+1} - \mathcal{K}^{t+1}\right). \tag{22}$$

Here t denotes the iteration index in the ADMM process.

In each iteration, the weights α_k are dynamically updated according to (12), ensuring that the optimization process adapts to the low-rank characteristics of different unfolding dimensions. It is worth emphasizing that the optimization variable $\mathcal{H}$ is always defined in the original Chla value space, with the observed entries constrained to match the incomplete input tensor $\mathcal{H}_0$. While the spatial, temporal, and periodic transforms $(\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3)$ are introduced to enhance low-rank structures and guide the optimization, the final output $\mathcal{H}^{t+1}$ directly represents the reconstructed Chla values. Therefore, no additional inverse transformation is required after optimization. For clarity, the entire completion and optimization procedure, including periodic restructuring, feature embedding, adaptive weighting, and ADMM updates, is summarized in Table S1.

III. Experiments and Analysis

A. Dataset and Study Area

The proposed method was tested using MODIS-Aqua daily Chla Level-3 data with a spatial resolution of 4 km, sourced from the National Aeronautics and Space Administration (NASA, https://oceancolor.gsfc.nasa.gov/l3/). In NASA satellite products, Level-3 (L3) data consist of geophysical retrievals that have been aggregated both spatially and temporally onto a standardized geo-referenced Earth grid for a specific timeframe [49]. For a value to be assigned to an L3 grid cell, at least one of the Level-2 (L2) geophysical data points must meet quality control criteria, be spatially located within the designated range of the L3 grid cell, and fall within the specified time window for aggregation. The quality control thresholds for generating L3 data are determined by the flags included in the ocean color L2 product, which are established during the NASA atmospheric correction process from Level-1B to Level-2.

The Bohai, Yellow, and East China Seas (24–41°N, 116–127°E) were selected as the experimental areas (Fig. 2). The Bohai and Yellow Seas are semi-enclosed oceans with pronounced intraannual fluctuations in Chla [50]. In contrast, the East China Sea is considered a semi-open ocean, and its Chla fluctuates relatively smoothly during the year. This variability facilitates the evaluation of the proposed method under different Chla patterns.

Within the experimental domain, we selected MODIS data from 2013 to 2023, which has an average missing rate of approximately 87% (Fig. 2). The missing rate is especially high in coastal areas; for example, more than 95% of the data is missing in the region near the mouth of the Yangtze River. In many Chla-related studies, regions within a certain range of the coastal zone were masked due to the large retrieval error and extremely high missing levels in these areas [23], [51]. Following this practice, we excluded regions with catastrophic levels of missing data (greater than 98%) from the accuracy assessment and spatiotemporal analyses.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

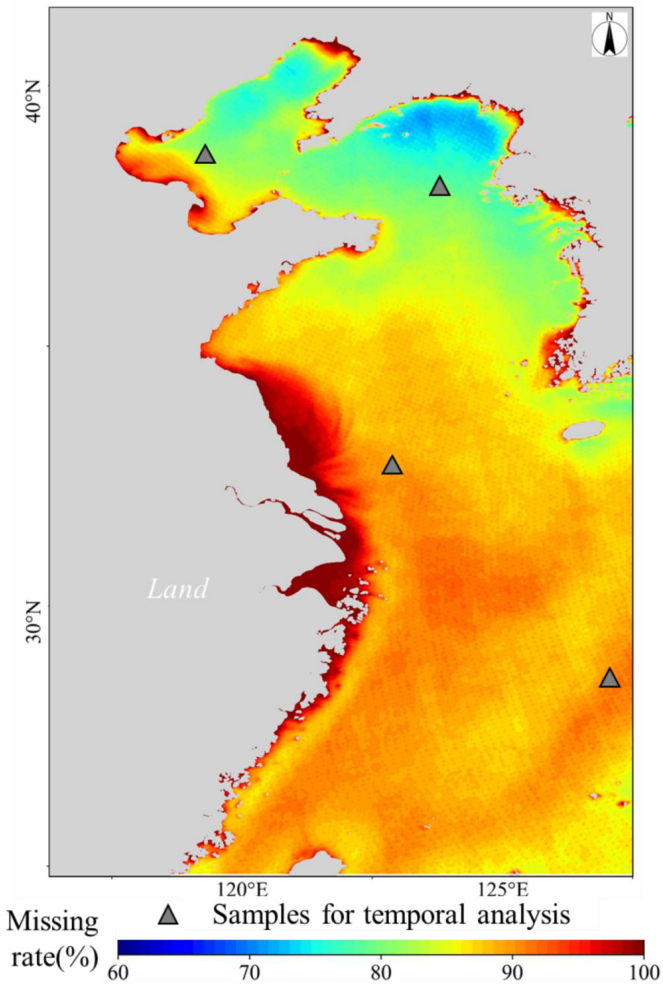


Fig. 2. Experimental area and the distribution of Chla data missing rates (2013–2023). Gray triangles mark the centers of subregions used for time-series analysis.

B. Performance Evaluation

1) Quantitative Validation With Synthetic Missing Data: To evaluate the proposed method under controlled conditions, a quantitative assessment was first conducted using synthetic gap scenarios. A total of 20,000 ocean pixels were randomly selected within the study area (Fig. S2), and 30% of valid observations in their Chla time series were randomly removed. The resulting missing rates ranged from 65% to 98%, covering a wide range of data loss situations (Fig. S3). The reconstruction accuracy was evaluated using three metrics: the correlation coefficient (CC), mean absolute percentage difference (MAPD), and root mean square difference (RMSD), defined as follows:

$$\mathbf{CC} = \frac{\operatorname{cov}\left(Y, \hat{Y}\right)}{\sqrt{\operatorname{var}(Y) \cdot \operatorname{var}\left(\hat{Y}\right)}} \tag{23}$$

$$\mathbf{MAPD} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{24}$$

$$\mathbf{RMSD} = \sqrt{\frac{1}{n} \sum\nolimits_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{25}$$

where Y represents the reconstructed Chla values, and $\hat{Y}$ denotes the observed Chla values from satellite retrievals. y_i and $\hat{y}_i$ correspond to the reconstructed and observed Chla of the ith sample, respectively. Since satellite-derived Chla data also contain retrieval uncertainties, the term "difference" rather than "error" is used to describe the deviations between reconstructed and observed Chla.

2) Validation With In Situ Observations: To further evaluate the real-world applicability of the method, independent validation was performed using in situ Chla measurements. These data were acquired from the Biological and Chemical Oceanography Data Management Office (BCO-DMO) (https://www.bcodmo.org/dataset/829141), collected during trans-Pacific voyages from Hong Kong to Long Beach (2009–2012) [52]. Because of the partial spatial and temporal overlap between these measurements and the primary MODIS-Aqua dataset, a separate validation experiment was conducted. The proposed method was applied to MODIS-Aqua Chla data corresponding to the in situ data time and space range, and the reconstructed values were compared against the in situ observations using CC, MAPD, and RMSD.

3) Qualitative Analysis of Reconstruction Performance: Qualitatively, pixels from several representative sea regions were selected to analyze the differences among Chla time series reconstructed by different methods. Additionally, the original and gap-filled Chla images were compared to evaluate the spatial recovery performance of each method.

4) Benchmark Methods for Comparison: To benchmark the proposed method, several widely used or representative gap-filling approaches were selected. These include linear interpolation, a simple and general method for time-series recovery; *DINEOF*, a widely adopted spatiotemporal interpolation method with parameters following [8]; *DINCAE*, a DL-based method that employs convolutional autoencoders for satellite data reconstruction [22]; CP decomposition-based tensor completion, which approximates the data tensor as a sum of rank-one components [39]; Tucker decomposition-based tensor completion, which captures low-rank structures using a core tensor and factor matrices [40]; and the climatological daily mean, which fills missing values using the multiyear average of Chla on the same calendar day, thereby leveraging seasonal periodicity but not accounting for interannual variability. All methods were directly applied to the daily Chla time series without auxiliary variables.

C. Experimental Settings

In the experiment, the 366th day was excluded from the MODIS time series during leap years to ensure consistent annual length. For the spatiotemporal feature embedding process, the parameters for transformation were set as follows after several experimental analyses (Fig. S4): in the Graph Laplacian transform, σ^2 and ϵ were set to 10^4 and 0.01, respectively; for the fractional difference transform, K and α were set to 3 and 0.98, respectively; in the periodic circulant transform, the period was fixed at 365 days. Within the tensor completion model, the initial weights of the three unfolding matrices were equally distributed ($\alpha_1 = \alpha_2 = \alpha_3 = 1/3$). In the ADMM algorithm, μ_t and the maximum number of iterations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

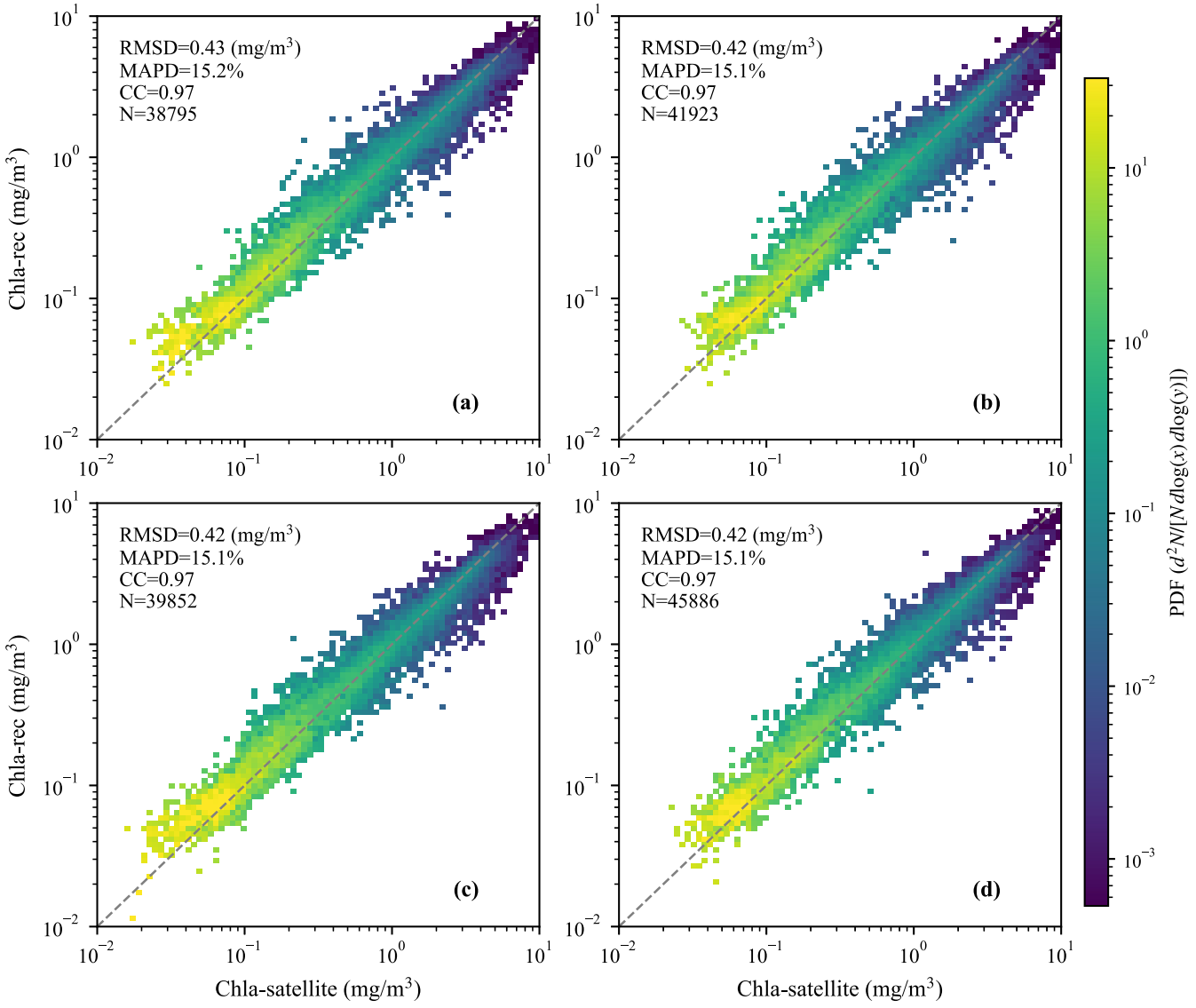


Fig. 3. Accuracy results of the proposed method for different years across the experimental area. (a) 2013. (b) 2016. (c) 2019. (d) 2022. The color indicates the joint probability density function (pdf) estimated in the log–log space, normalized by the total number of samples and bin area.

were set to 0.01 and 50, respectively, to ensure computational efficiency and convergence.

D. Experimental Results

1) Accuracy Evaluation:

a) Synthetic data results: The accuracy of the proposed method was validated across the entire study area over multiple years (2013, 2016, 2019, and 2022). Fig. 3 illustrates the performance of the proposed method throughout these years, achieving CC values of about 0.97. Additionally, RMSDs are consistently below 0.43 mg/m^3, and MAPDs are below 15.2%. These results demonstrate robust model performance across different years. Although the validation points had different random missing data in each year, the accuracy differences were negligible, showing the excellent reconstruction ability and robustness of the proposed method. Moreover, despite the different concentration ranges of Chla among years, the reconstructed and original values are symmetrically distributed around the 1:1 line.

To evaluate the method under different missing rates, we calculated accuracy metrics after grouping validation pixels by missing ratio. As shown in Fig. 4(a)–(c), most pixels fall in the 85%–95% missing range, with fewer pixels at 65%–75%, consistent with the overall data status in the region. When the missing rate is 65%–75%, the proposed method achieved the best reconstruction (CC = 0.97, RMSD = 0.41 mg/m^3, MAPD = 14.1%). At 75%–85%, the metrics were CC = 0.97, RMSD = 0.42 mg/m^3, and MAPD = 14.1%; at 85%–95%, CC = 0.96, RMSD = 0.44 mg/m^3, and MAPD = 15.3%.

As shown in Fig. 4(d)–(f), the accuracy metrics exhibit a clear decreasing trend with increasing missing rates (5% intervals). Nevertheless, the model maintains acceptable accuracy under high missing ratios: when missing rates exceed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

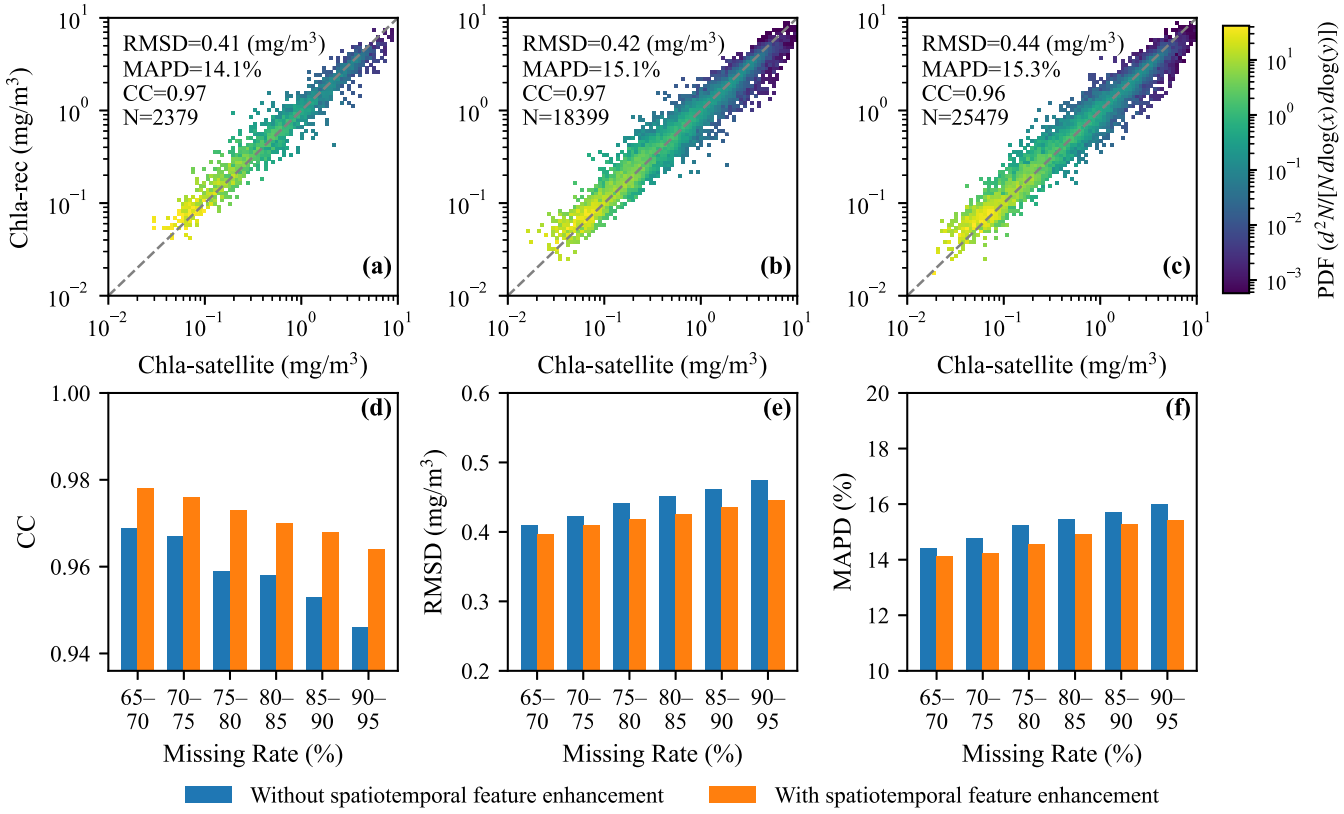


Fig. 4. Variation of the accuracy of the proposed method under different missing-rate conditions. (a)–(c) Density heatmaps showing the statistical relationship between satellite-derived and reconstructed Chla for missing-rate intervals of 65%–75%, 75%–85%, and 85%–95%, respectively. The color indicates the joint pdf estimated in the log–log space, normalized by the total number of samples and bin area. The corresponding performance metrics (CC, RMSD, MAPD, and N) are annotated in each panel. (d)–(f) Grouped bar charts summarizing reconstruction performance across different missing-rate intervals: (d) CC, (e) RMSD, and (f) MAPD.

85%, RMSD and MAPD remain 0.43 and 15.2%, respectively; even at 95%, RMSD = 0.44 and MAPD = 15.4%. Only at extreme rates (98%) does recovery become unstable and highly dependent on the spatial–temporal distribution of observations. Thus, 95% can be considered the practical upper bound for reliable reconstruction.

Furthermore, we compared the performance with and without spatiotemporal feature embedding. Results show that although accuracy slightly decreases without embedding (minimum CC = 0.94, maximum RMSD = 0.47), performance remains acceptable. This demonstrates the contribution of embedding and the reliability of the proposed approach for highly incomplete daily Chla time series.

b) In situ data results: The comparison between reconstructed Chla values and in situ measurements from the BCO-DMO dataset is shown in Fig. 5, together with valid satellite-derived Chla for reference. All values were $\log_{10}$-transformed to reduce the influence of extreme values and improve interpretability. The reconstructed values (blue triangles) exhibit strong consistency with in situ data (CC = 0.71, RMSD = 0.64 mg/m^3), while valid satellite observations (red squares) performed slightly worse (CC = 0.63, RMSD = 0.69 mg/m^3). This indicates that the reconstruction not only provides reliable estimates in missing regions but also reduces noise and retrieval artifacts inherent in single-pass satellite products. We note that MAPD can be relatively large because it is a percentage-based metric and is sensitive to low in

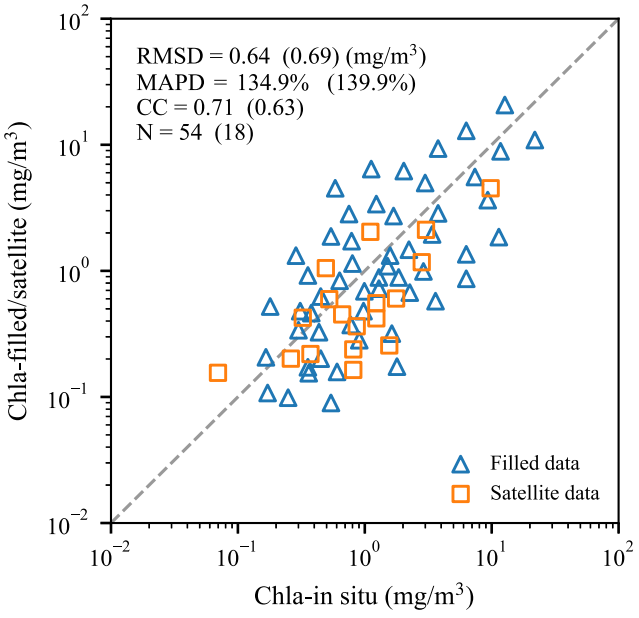


Fig. 5. Comparison of in situ Chla with reconstructed values (blue triangles) and valid satellite observations (orange squares).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISON OF ACCURACY FOR DIFFERENT METHODS USING SYNTHETIC DATA IN 2022

Method	CC	RMSD (mg/m^3)	MAPD
Climatological daily mean	0.63	0.84	43.8%
Linear Interpolation	0.76	0.76	40.9%
DINEOF	0.82	0.64	29.1%
DINCAE	0.84	0.57	24.8%
CP	0.86	0.58	23.4%
Tucker	0.87	0.56	20.3%
Proposed Method	**0.97**	**0.42**	**15.1%**

situ Chla values (i.e., small denominators), such that modest absolute deviations may translate into large relative deviations. In addition, representativeness differences between point-wise in situ samples and satellite pixel-scale estimates may further contribute to the percentage deviation. Despite the sensitivity of MAPD to low in situ concentrations (approximately 60% of the in situ Chla measurements are below 1 mg/m^3), the relatively high CC together with the moderate RMSD suggests that the reconstructed Chla captures the overall variability and achieves a reasonable absolute agreement with in situ observations, supporting the robustness of the proposed method under real in situ validation conditions.

c) Baseline comparison: To quantitatively assess the reconstruction performance of the proposed method, we compared it with baseline methods on the 2022 daily Chla dataset with artificially introduced missing values. As shown in Table I, the proposed method achieved the highest accuracy across all evaluation metrics (CC = 0.97, RMSD = 0.42 mg/m^3, MAPD = 15.1%), substantially outperforming the next-best method (Tucker), which yielded CC = 0.87 and MAPD = 20.3%.

Among the simple baselines, climatological daily mean and linear interpolation provided only limited performance. The climatological approach, which fills gaps with the multiyear daily mean, could reproduce the broad seasonal cycle but failed to capture interannual anomalies or short-term variability (Fig. S5), resulting in low correlation (CC <0.70) and high errors (RMSD >0.80 mg/m^3, MAPD >43.0%). Linear interpolation performed slightly better (CC = 0.76, RMSD = 0.76 mg/m^3, MAPD = 40.9%), but its accuracy also degraded severely under high missingness. Similarly, DINEOF reduced errors compared with climatology and linear interpolation, but still remained limited (MAPD = 29.1%).

In contrast, DL-based DINCAE and low-rank tensor models such as CP and Tucker performed relatively better, but they were still outperformed by the proposed method. These results highlight the effectiveness of integrating spatial, temporal, and periodic transformations prior to tensor completion.

2) Temporal Performance After Gap-Filling: To evaluate reconstruction performance under varying marine environments, four representative subregions were selected, including an enclosed bay, a semi-enclosed bay, a nearshore region, and an offshore region. Fig. 6 presents both the daily and monthly reconstructed Chla time series within these areas. The quantitative metrics shown are used to indicate the consistency between reconstructed and original observations at the site level, while the overall performance assessment relies on large-sample statistics.

Because of the high missing rates, linear interpolation was found unsuitable for reliable reconstruction and is therefore excluded from the visual comparison. Across all subregions, the proposed method exhibits strong temporal coherence and stability, effectively preserving both short-term variability and seasonal evolution of Chla. For example, in the enclosed bay [Fig. 6(a)], the proposed method achieves a high agreement with observations (CC = 0.89) with relatively small errors (RMSD = 0.54 mg/m^3, MAPD = 4.7%), while maintaining Chla fluctuations. In the nearshore region [Fig. 6(c)], where observations are highly variable, the proposed method still tracks the temporal dynamics well (CC = 0.95, RMSD = 0.3 mg/m^3, MAPD = 17.2%) and effectively suppresses high-frequency noise without distorting the seasonal pattern.

At the monthly scale [Fig. 6(b), (d), (f), and (h)], the proposed method continues to produce stable seasonal trajectories across different environments. It is also observed that the monthly-scale accuracy metrics were lower than those at the daily scale. This is mainly because monthly results contain far fewer valid samples, and the monthly observations are aggregated from sparse and unevenly distributed daily measurements, which may introduce temporal representativeness errors and reduce the effective dynamic range, thereby making CC/RMSD/MAPD less stable and sometimes less favorable.

In comparison, DINEOF generally performs reasonably, but in regions with uneven observation distributions [e.g., Fig. 6(a) and (e) and (c) and (g)], its results fluctuate more strongly, likely due to its sensitivity to localized outliers and reliance on statistical assumptions in spatial reconstruction. DINCAE produces relatively smooth curves but does not show substantial advantages over traditional methods in the selected regions. CP and Tucker decomposition-based methods capture seasonal trends to some extent, but as observed in Fig. 6(b) and (f), they tend to flatten seasonal amplitudes and suppress peaks, particularly in offshore regions where Chla variability is relatively low. These results demonstrate that the proposed method consistently outperforms alternative approaches in maintaining temporal dynamics, suppressing noise, and achieving stable reconstructions across both daily and monthly scales in diverse marine environments.

3) Spatial Performance After Gap-Filling: Fig. 7 illustrates the spatial distribution of Chla reconstruction results and the corresponding MAPD under three different missing data scenarios. The original Chla images with missing regions are shown in Fig. 7(a7), (b7), and (c7), where gaps are mainly distributed in the Bohai and Yellow Seas Fig. 7(a7), central offshore regions Fig. 7(b7), and the East China Sea Fig. 7(c7).

Among all methods, linear interpolation [Fig. 7(a1), (b1), and (c1)] shows the poorest performance, with clear overestimation in the Bohai Sea and blocky noise in regions with continuous gaps, as further confirmed by high MAPD values in Fig. 7(d1). DINEOF and DINCAE [Fig. 7(a2), (a3), (b2), (b3), (c2), and (c3)] produce smoother results with fewer artifacts, but both tend to oversmooth spatial variability, especially near estuaries and coastal zones, leading to noticeable errors in their MAPD maps [Fig. 7(d2) and 7(d3)]. Low-rank

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

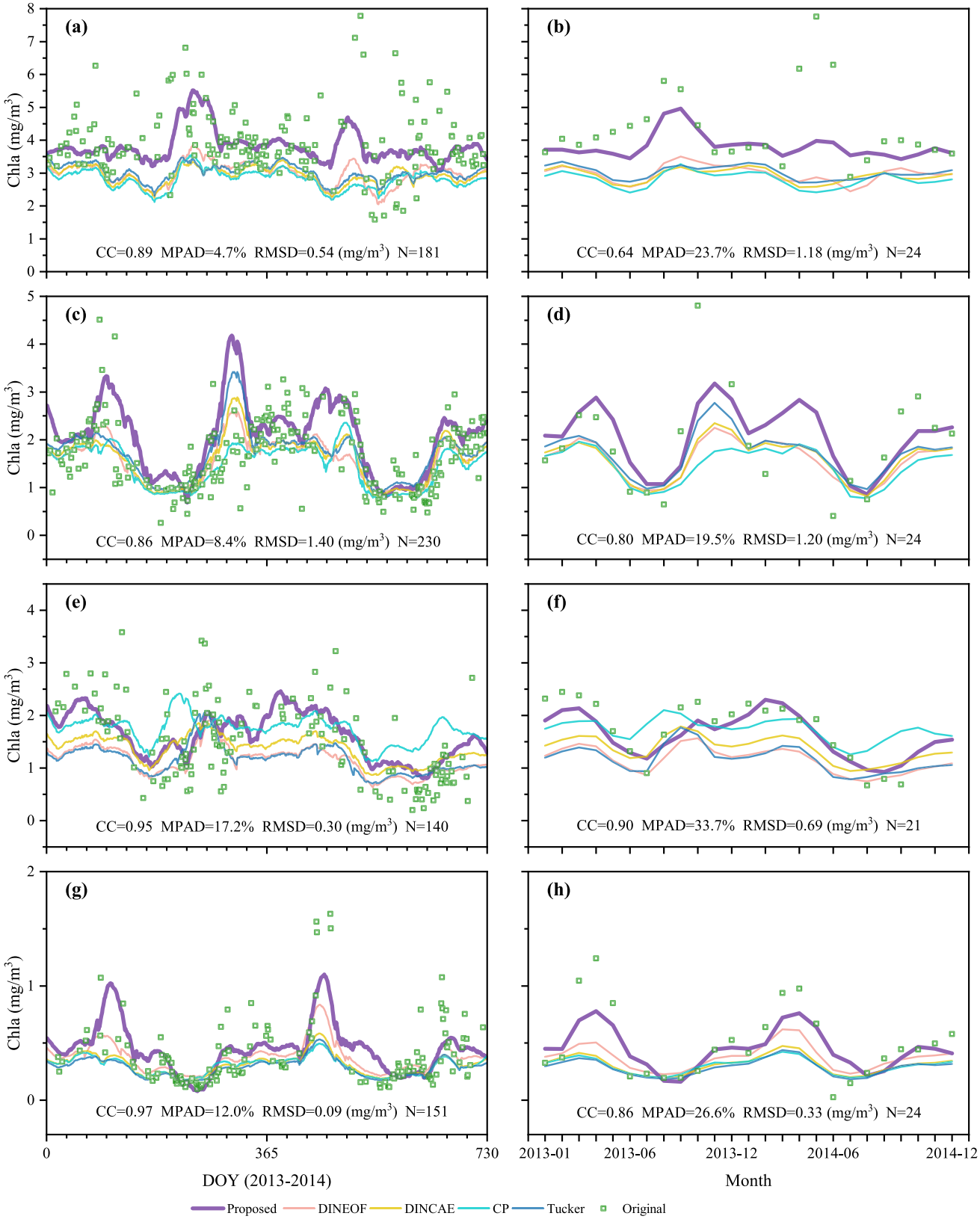


Fig. 6. (a), (c), (e), and (g) Daily and (b), (d), (f), and (h) monthly gap-filled Chla time series of different methods at four representative subregions (each sized 10×10 pixels). (a) and (b) Enclosed bay. (c) and (d) Semi-enclosed bay. (e) and (f) Nearshore. (g) and (h) Offshore. Accuracy metrics (CC, RMSD, and MAPD) are computed between the reconstructed Chla of the proposed method and the original observations at time steps where observations are available. These metrics are reported as indicative site-level measures, given the limited sample size and irregular temporal sampling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

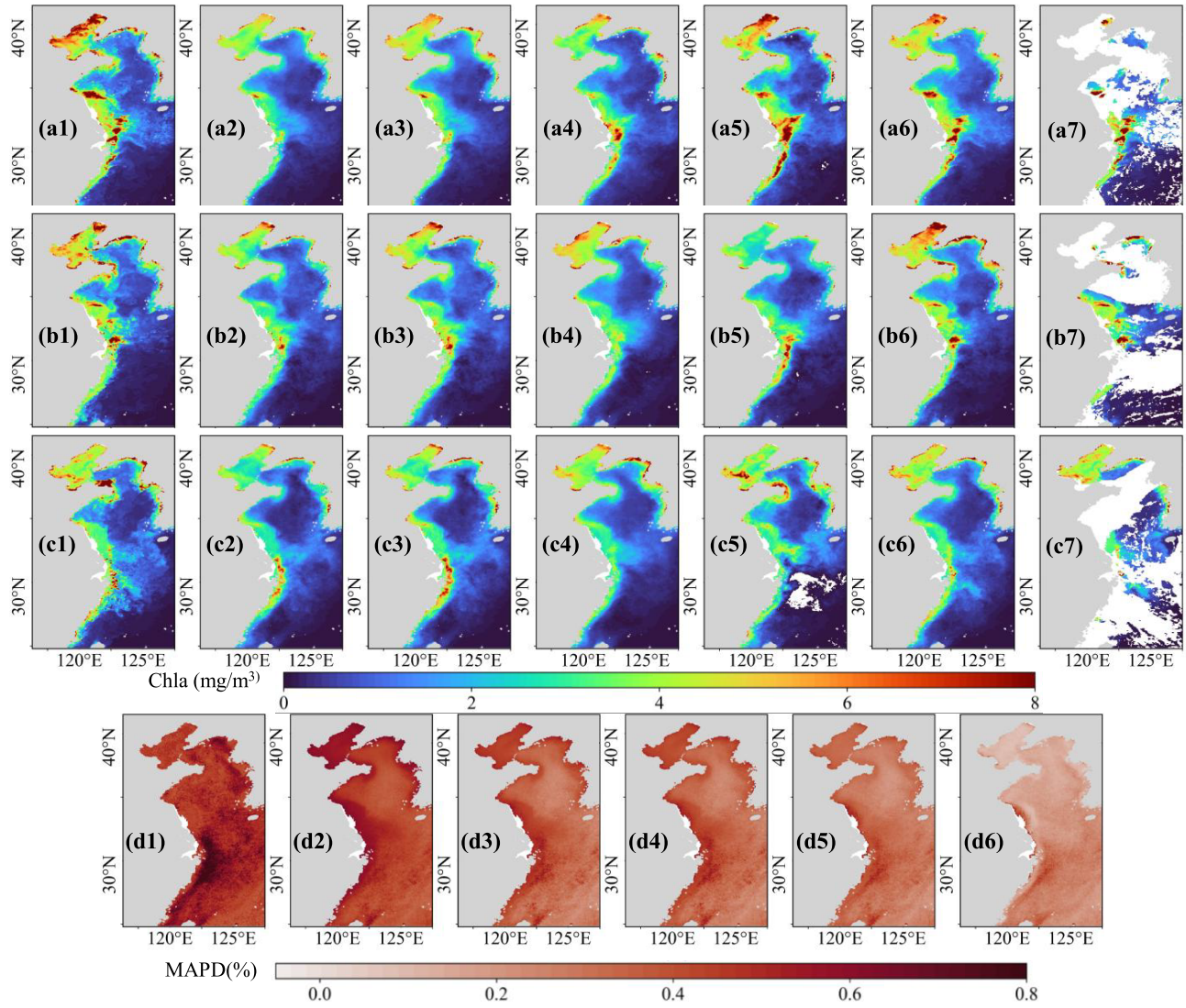


Fig. 7. Spatial distribution of Chla after gap-filling and corresponding MAPD for different reconstruction methods under various missing data scenarios. (a1), (b1), (c1), and (d1) Linear interpolation. (a2), (b2), (c2), and (d2) DINEOF. (a3), (b3), (c3), and (d3) DINCAE. (a4), (b4), (c4), and (d4) CP. (a5), (b5), (c5), and (d5) Tucker. (a6), (b6), (c6), and (d6) Proposed method. (a7), (b7), and (c7) Raw Chla images with different missing patterns.

decomposition methods, including CP and Tucker [Fig. 7(a4), (a5), (b4), (b5), (c4), and (c5)], generally preserve large-scale patterns but struggle with fine-scale details and often introduce excessive smoothness in heterogeneous regions, as indicated by moderate MAPD levels [Fig. 7(d4) and (d5)].

In contrast, the proposed method [Fig. 7(a6), (b6), and (c6)] achieves the most accurate and spatially consistent reconstructions, effectively recovering both large-scale patterns and local variations under different missing scenarios. It avoids overestimation in high-Chla areas, restores nearshore and estuarine details with high fidelity, and ensures smooth transitions in offshore regions. The superiority of the proposed method is also reflected in the lowest MAPD values [Fig. 7(d6)], confirming its robustness in handling severe and spatially diverse data gaps. Furthermore, to assess temporal consistency of the reconstructed results, Fig. S6 presents examples of consecutive daily Chla maps before and after reconstruction, confirming that the proposed method produces stable spatial patterns without abrupt day-to-day variations often found in existing interpolated datasets.

TABLE II
ABLATION EXPERIMENT RESULTS

Variant	CC	RMSD (mg/m^3)	MAPD
Full model (all modules)	0.97	0.42	15.1%
w/o Graph Laplacian transform	0.87	0.56	22.7%
w/o Fractional difference transform	0.89	0.54	21.6%
w/o Periodic circulant transform	0.90	0.50	18.1%

IV. DISCUSSION

A. Ablation and Efficiency Analysis

To comprehensively evaluate the effectiveness and practicality of the proposed method, we conducted an ablation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

study by selectively removing each of the three key transform modules: graph-based spatial smoothing, fractional temporal differencing, and the periodic circulant constraint (Table II). Among these components, the graph Laplacian transform had the most significant impact on reconstruction performance. Its removal resulted in a substantial drop in CC (from 0.97 to 0.87), an increase in RMSD from 0.42 to 0.56 mg/m^3, and a notable rise in MAPD from 15.1% to 22.7%. These results indicate that spatial structure preservation is essential for accurate gap-filling, especially in areas with high spatial variability. The fractional differencing module also contributed positively, with its exclusion leading to a MAPD increase of over 6%. Although the periodic circulant transform had a comparatively smaller impact, it still led to a measurable degradation in performance (MAPD increased from 15.1% to 18.1%) and is therefore important for capturing seasonal continuity across years. These results confirm the complementary contributions of the three modules in capturing spatial structure, temporal smoothness, and seasonal recurrence.

We further compared the practical efficiency of our method with several representative approaches using the same dataset: a 10-year daily Chla time series over 50 spatial pixels. All experiments were conducted on a Windows PC equipped with an Intel Core i9-10900 processor and 32 GB of RAM. The proposed method completed the full reconstruction in approximately 9 min, which is faster than DINEOF (14 min) and DINCAE (13 min), and slightly faster than Tucker decomposition (11 min) and CP decomposition (10 min), as shown in Supplementary Table S1. Unlike DINCAE, which requires a dedicated training phase and is sensitive to model configuration, our method is fully unsupervised and model-free, avoiding training overhead while maintaining competitive accuracy. Although linear interpolation was the fastest, it performs poorly in complex scenarios. Overall, the proposed method achieves a stable balance between performance and efficiency for gap-filling in remote sensing Chla time series with severe missingness.

B. Advantage of the Proposed Method

The proposed spatiotemporal feature-enhanced tensor completion method provides a robust and general solution for reconstructing daily ocean color data in the presence of severe missingness. By explicitly incorporating spatial, temporal, and seasonal priors through graph Laplacian, fractional difference, and periodic circular transforms, the method effectively enhances the intrinsic coherence of the Chla time-series tensor before applying low-rank completion. This feature-aware enhancement allows the model to preserve both large-scale variability and short-term dynamics, addressing the oversmoothing issue common in conventional tensor decompositions. As demonstrated by multiyear synthetic evaluations, the proposed method consistently achieves high reconstruction accuracy across varying years (CC >0.97, RMSD <0.43 mg/m^3) and remains stable under high missing rates. Even when up to 95% of data are missing, it maintains a CC of 0.96 and a MAPD below 15.5% (Fig. 4). Furthermore, the agreement with in situ observations (CC = 0.71, RMSD = 0.64 mg/m^3) confirms its effectiveness under real-world oceanographic conditions, despite differences in spatial resolution and measurement uncertainty. These results are comparable to or better than those reported in previous studies using DINEOF and machine learning approaches for Chla reconstruction [12], [22].

In addition to accuracy performance, the proposed method demonstrates advantages over existing approaches in terms of adaptability and generalization. It performs consistently across diverse marine environments, from enclosed bays to offshore regions, and reconstructs both random and spatially contiguous gaps. Compared with classical tensor decomposition methods, it reduces reconstruction errors by over 5 percentage points relative to Tucker (20.3%) and by more than 14 points compared to DINEOF (29.1%) (Table I), highlighting its superior capability in handling extreme missingness. Unlike interpolation-based approaches prone to spatial artifacts and DL models that require large-scale supervised training, the proposed method provides an unsupervised, training-free alternative that generalizes well across scenes. Its feature-enhanced low-rank modeling substantially improves the expressiveness of the tensor representation while avoiding oversmoothing. Visual comparisons further show that the method better recovers temporal dynamics and spatial patterns, particularly in high-variability areas (Figs. 6 and 7). These advantages make the proposed method suitable for long-term Chla reconstruction and extensible to other ocean color variables (e.g., Kd_{490}) and environmental datasets with severe data gaps, supporting operational marine monitoring and climate-related analyses.

C. Limitations and Future Works

While the proposed method performs well across a range of scenarios, certain limitations remain. One issue lies in its applicability to regions with near-complete data loss. Our experiments demonstrated that the method remains effective up to a missing data ratio of about 95%, but beyond this level, such as 98%, the reconstruction becomes unstable and highly dependent on the distribution of the remaining observed entries. These areas are typically found in coastal zones, where water optical properties are complex and Chla retrieval is already subject to high uncertainty. In such cases, the limited number of valid observations over multiyear periods (e.g., fewer than 37 observations over five years) makes it difficult to extract reliable temporal features. These regions were typically excluded from marine ecological change analyses [23] and thus masked in the present study. Although they represent only a small proportion ($\approx$ 1.1%) of the total study area, future research could explore the integration of ancillary datasets, such as sea surface temperature, wind speed, or turbidity, to support reconstruction in these challenging areas using advanced learning or hybrid modeling approaches.

Another consideration is the tradeoff between spatiotemporal scale and computational feasibility. Although tensor completion methods can theoretically accommodate large-scale data, applying them to large-area, long-term daily datasets requires considerable memory and processing time. Conversely, when spatial or temporal coverage is too limited, especially under high missing rates, the low-rank assumption

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

may not hold sufficiently to enable reliable reconstruction. Future work may investigate scalable implementation strategies such as chunk-based processing and adaptive spatial-temporal window selection to improve applicability to global-scale ocean color datasets. In addition, it is of interest to assess the adaptability of the proposed framework to other ocean color products, such as the diffuse attenuation coefficient at 490 nm and particulate organic carbon, which share similar spatiotemporal patterns and face comparable data quality issues.

In addition to methodological considerations, the reconstructed daily Chla dataset offers opportunities for more process-oriented applications. By filling large data gaps, the dataset enables more rational analyses of phytoplankton bloom dynamics, interannual anomalies, and extreme events that are often obscured in raw satellite observations. Such process-oriented investigations have the potential to reveal new patterns of Chla variability and provide valuable ecological insights, which will be an important direction of future work.

Furthermore, it should be noted that in our earlier study, we developed a neighborhood-based spatiotemporal gap-filling (STGF) method [18], which reconstructs missing values by identifying neighboring pixels with similar temporal trajectories. This approach is effective under moderate missing conditions because it relies on climatological mean series derived from multiple years of valid data to guide the similarity search. However, when missingness becomes severe, such climatologies are highly uncertain, limiting the applicability of STGF. In contrast, the tensor-based framework presented in this study incorporates global multidimensional spatiotemporal priors and adaptive low-rank constraints, allowing it to achieve robust performance. Future research may combine these strategies, using tensor completion for global stability and STGF for local refinement, to further enhance reconstruction quality under diverse conditions.

V. Conclusion

The inevitable gaps in Chla time-series data substantially hinder their application, especially for daily products with high missing rates, making gap-filling a challenging task. This study proposes a spatiotemporal tensor-based gap-filling method to address large gaps in daily Chla time-series products. By rearranging the Chla tensor and applying three feature extraction transforms, including Graph Laplacian, fractional difference, and periodic circulant transforms, the method effectively incorporates spatial, temporal, and seasonal dependencies into an adaptive weighted low-rank tensor completion framework. Experimental evaluations across multiple years (2013, 2016, 2019, and 2022) demonstrate that the proposed method maintains stable and high reconstruction accuracy, with CC consistently exceeding 0.9, MAPDs lower than 14.2%, and RMSDs less than 0.43 mg/m^3. The model also performs robustly under varying missing rates (65%–95%), including extreme conditions with 95% data loss. In comparative experiments, the proposed method outperforms commonly used interpolation and tensor-based approaches, including linear interpolation, DINEOF, DINCAE, CP, and Tucker, both in quantitative metrics and in its ability to preserve temporal dynamics and spatial patterns. Unlike DL approaches, the method does not require external training data or auxiliary variables, making it highly adaptable to diverse ocean regions. Overall, the proposed method provides an effective and generalizable framework for reconstructing daily ocean color time series, offering practical support for the development of long-term, gap-free Chla datasets and advancing marine ecosystem studies.

References

[1] H. Demarcq, G. Reygondeau, S. Alvain, and V. Vantrepotte, "Monitoring marine phytoplankton seasonality from space," *Remote Sens. Environ.*, vol. 117, pp. 211–222, Feb. 2012.

[2] D. A. Siegel et al., "Regional to global assessments of phytoplankton dynamics from the SeaWiFS mission," *Remote Sens. Environ.*, vol. 135, pp. 77–91, Aug. 2013.

[3] Y. Bai et al., "A mechanistic semi-analytical method for remotely sensing sea surface pCO_2 in river-dominated coastal oceans: A case study from theEastChinaSea," *J. Geophys. Res., Oceans*, vol. 120, no. 3, pp. 2331–2349, Mar. 2015.

[4] G. Jiang, S. A. Loiselle, D. Yang, R. Ma, W. Su, and C. Gao, "Remote estimation of chlorophyll a concentrations over a wide range of optical conditions based on water classification from VIIRS observations," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111735.

[5] B. A. Franz et al., "Nasa multi-mission ocean color reprocessing 2018.0," in *Proc. 24th Ocean Opt.*, Dubrovnik, Croatia, 2018, pp. 8–12.

[6] C. Hu, L. Feng, and Q. Guan, "A machine learning approach to estimate surface chlorophyll a concentrations in global oceans from satellite measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4590–4607, Jun. 2021.

[7] N. Pahlevan et al., "Seamless retrievals of chlorophyll—A from Sentinel-2 (MSI) and sentinel-3 (OLCI) in inland and coastal waters: A machine-learning approach," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111604.

[8] X. Liu and M. Wang, "Gap filling of missing data for VIIRS global ocean color products using the DINEOF method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4464–4476, Aug. 2018.

[9] L. Ćatipović, F. Matić, and H. Kalinić, "Reconstruction methods in oceanographic satellite data observation—A survey," *J. Mar. Sci. Eng.*, vol. 11, no. 2, p. 340, Feb. 2023.

[10] R. He, R. H. Weisberg, H. Zhang, F. E. Müller-Karger, and R. W. Helber, "A cloud-free, satellite-derived, sea surface temperature analysis for the west Florida shelf," *Geophys. Res. Lett.*, vol. 30, no. 15, Aug. 2003, Art. no. 1811.

[11] M. Vargas, C. W. Brown, and M. R. P. Sapiano, "Phenology of marine phytoplankton from satellite ocean color measurements," *Geophys. Res. Lett.*, vol. 36, no. 1, p. 01608, Jan. 2009.

[12] M. Xing, F. Yao, J. Zhang, X. Meng, L. Jiang, and Y. Bao, "Data reconstruction of daily MODIS chlorophyll–A concentration and spatio-temporal variations in the Northwestern Pacific," *Sci. Total Environ.*, vol. 843, Oct. 2022, Art. no. 156981.

[13] A. Alvera-Azcárate, A. Barth, M. Rixen, and J. M. Beckers, "Reconstruction of incomplete oceanographic data sets using empirical orthogonal functions: Application to the Adriatic Sea surface temperature," *Ocean Model.*, vol. 9, no. 4, pp. 325–346, 2005.

[14] Y. Wang and D. Liu, "Reconstruction of satellite chlorophyll—A data using a modified DINEOF method: A case study in the Bohai and Yellow Seas, China," *Int. J. Remote Sens.*, vol. 35, no. 1, pp. 204–217, Jan. 2014.

[15] A. Alvera-Azcárate, A. Barth, G. Parard, and J.-M. Beckers, "Analysis of SMOS sea surface salinity data using DINEOF," *Remote Sens. Environ.*, vol. 180, pp. 137–145, Jul. 2016.

[16] X. Liu and M. Wang, "Global daily gap-free ocean color products from multi-satellite measurements," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102714.

[17] A. Barth, A. Alvera-Azcárate, M. Licer, and J.-M. Beckers, "DINCAE 1.0: A convolutional neural network with error estimates to reconstruct sea surface temperature satellite observations," *Geoscientific Model Develop.*, vol. 13, no. 3, pp. 1609–1622, Mar. 2020.

[18] G. Zhou, M. Liu, L. Xu, and L. Li, "A gap-filling method for satellite-derived chlorophyll—A time series based on neighborhood spatiotemporal information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103724.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.

[19] M. Jouini, M. Lévy, M. Crépon, and S. Thiria, "Reconstruction of satellite chlorophyll images under heavy cloud coverage using a neural classification method," *Remote Sens. Environ.*, vol. 131, pp. 232–246, Apr. 2013.

[20] S. Chen, C. Hu, B. B. Barnes, Y. Xie, G. Lin, and Z. Qiu, "Improving ocean color data coverage through machine learning," *Remote Sens. Environ.*, vol. 222, pp. 286–302, Mar. 2019.

[21] A. Barth, A. Alvera-Azcárate, C. Troupin, and J.-M. Beckers, "DINCAE 2.0: Multivariate convolutional neural network with error estimates to reconstruct sea surface temperature satellite and altimetry observations," *Geosci. Model Develop.*, vol. 15, no. 5, pp. 2183–2196, Mar. 2022.

[22] X. Luo, J. Song, J. Guo, Y. Fu, L. Wang, and Y. Cai, "Reconstruction of chlorophyll–A satellite data in Bohai and Yellow Sea based on DINCAE method," *Int. J. Remote Sens.*, vol. 43, no. 9, pp. 3336–3358, May 2022.

[23] Z. Hong et al., "A global daily gap-filled chlorophyll—A dataset in open oceans during 2001–2021 from multisource information using convolutional neural networks," *Earth Syst. Sci. Data*, vol. 15, no. 12, pp. 5281–5300, Nov. 2023.

[24] M. Ye, B. Li, J. Nie, Y. Qian, and L.-L. Yang, "Ca-STANet: Spatiotemporal attention network for chlorophyll—A prediction with gap-filled remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4203314.

[25] E. Jang, D. Han, J. Im, T. Sung, and Y. Jun Kim, "Deep learning-based gap filling for near real-time seamless daily global sea surface salinity using satellite observations," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104029.

[26] H. Wang, Y. Zhou, and X. Li, "GDCM: Generalized data completion model for satellite observations," *Remote Sens. Environ.*, vol. 324, Jul. 2025, Art. no. 114760.

[27] F. Zhang, H. Kung, F. Zhang, C. Yang, and J. Gan, "AI-powered spatiotemporal imputation and prediction of chlorophyll—A concentration in coastal ecosystems," *Nature Commun.*, vol. 16, no. 1, p. 7656, Aug. 2025.

[28] Y. Yang, H. Wang, and X. Li, "Deep learning for enhanced ocean color remote sensing: A foundation model approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4210718.

[29] T. Lin, Q. He, W. Zhan, and H. Zhan, "Persistent data gap in ocean color observations over the East China Sea in winter: Causes and reconstructions," *Remote Sens. Lett.*, vol. 11, no. 7, pp. 667–676, Jul. 2020.

[30] C. Ma, J. Zhao, B. Ai, and S. Sun, "Two-decade variability of sea surface temperature and chlorophyll—A in the northern South China Sea as revealed by reconstructed cloud-free satellite data," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9033–9046, Nov. 2021.

[31] T. Wang et al., "Revisiting the intraseasonal variability of chlorophyll—A in the adjacent luzon strait with a new gap-filled remote sensing data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4201311.

[32] W. Xie, T. Wang, and W. Jiang, "Spatio-temporal variability of surface chlorophyll a in the Yellow Sea and the East China Sea based on reconstructions of satellite data of 2001–2020," *J. Oceanol. Limnol.*, vol. 42, no. 2, pp. 390–407, Dec. 2023.

[33] A. M. Kuhn et al., "Temporal and spatial scales of correlation in marine phytoplankton communities," *J. Geophys. Res., Oceans*, vol. 124, no. 12, pp. 9417–9438, Dec. 2019.

[34] C. Jayaram, N. Priyadarshi, J. P. Kumar, T. V. S. U. Bhaskar, D. Raju, and A. J. Kochuparampil, "Analysis of gap-free chlorophyll—A data from MODIS in Arabian Sea, reconstructed using DINEOF," *Int. J. Remote Sens.*, vol. 39, no. 21, pp. 7506–7522, Nov. 2018.

[35] M. K.-P. Ng, Q. Yuan, L. Yan, and J. Sun, "An adaptive weighted tensor completion method for the recovery of remote sensing images with missing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3367–3381, Jun. 2017.

[36] D. Chu et al., "Long time-series NDVI reconstruction in cloud-prone regions via spatio-temporal tensor completion," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112632.

[37] T.-Y. Ji, N. Yokoya, X. X. Zhu, and T.-Z. Huang, "Nonlocal tensor completion for multitemporal remotely sensed images' inpainting," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3047–3061, Jun. 2018.

[38] G. Zheng and X. Li, "Empirical function method: A precise approach for filling data gaps in satellite sea surface temperature imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4200114.

[39] H. He, J. Yan, L. Wang, D. Liang, J. Peng, and C. Li, "Bayesian temporal tensor factorization-based interpolation for time-series remote sensing data with large-area missing observations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408013.

[40] Y. Liu, Z. Long, H. Huang, and C. Zhu, "Low CP rank and tucker rank tensor completion for estimating missing components in image data," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 4, pp. 944–954, Apr. 2020.

[41] H. Shu, H. Wang, J. Peng, and D. Meng, "Low-rank tensor completion with 3-D spatiotemporal transform for traffic data imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 1–15, Nov. 2024.

[42] E. Candès and B. Recht, "Exact matrix completion via convex optimization," *Commun. ACM*, vol. 55, no. 6, pp. 111–119, Jun. 2012.

[43] S. Gandy, B. Recht, and I. Yamada, "Tensor completion and low-n-rank tensor recovery via convex optimization," *Inverse Problems*, vol. 27, no. 2, Feb. 2011, Art. no. 025010.

[44] J. Liu, P. Musialski, P. Wonka, and J. Ye, "Tensor completion for estimating missing values in visual data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 208–220, Jan. 2013.

[45] V. Satopaa, J. Albrecht, D. Irwin, and B. Raghavan, "Finding a 'kneedle' in a haystack: Detecting knee points in system behavior," in *Proc. 31st Int. Conf. Distrib. Comput. Syst. Workshops*, Jun. 2011, pp. 166–171.

[46] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[47] J. Bai, J. Li, F. Xu, and H. Zhang, "Generalized symmetric ADMM for separable convex optimization," *Comput. Optim. Appl.*, vol. 70, no. 1, pp. 129–170, May 2018.

[48] C. Ling, G. Yu, L. Qi, and Y. Xu, "T-product factorization method for internet traffic data completion with spatio-temporal regularization," *Comput. Optim. Appl.*, vol. 80, no. 3, pp. 883–913, Dec. 2021.

[49] J. P. Scott and P. J. Werdell, "Comparing level-2 and level-3 satellite ocean color retrieval validation methodologies," *Opt. Exp.*, vol. 27, no. 21, pp. 30140–30157, 2019.

[50] Y. Fu, S. Xu, and J. Liu, "Temporal–spatial variations and developing trends of chlorophyll—A in the Bohai Sea, China," *Estuarine, Coastal Shelf Sci.*, vol. 173, pp. 49–56, May 2016.

[51] D. Righetti, M. Vogt, N. Gruber, A. Psomas, and N. E. Zimmermann, "Global pattern of phytoplankton diversity driven by temperature and environmental variability," *Sci. Adv.*, vol. 5, no. 5, p. 6253, May 2019.

[52] S. Clayton, H. I. Palevsky, L. Thompson, and P. D. Quay, "Synoptic mesoscale to basin scale variability in biological productivity and chlorophyll in the Kuroshio extension region," *J. Geophys. Res., Oceans*, vol. 126, no. 11, Nov. 2021, Art. no. e2021JC017782.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:57:08 UTC from IEEE Xplore. Restrictions apply.